

Machine Learning Models for Advanced Air Quality Prediction

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ABSTRACT

Air quality stands as a pivotal factor influencing public health and well-being, shaping urban planning strategies, health management practices, and environmental policies. Having meticulously scrutinized PM2.5 concentration data from 42 monitoring stations across Bishkek, we have employed meteorological factors, including temperature and humidity from these stations to model air quality. We have thus implemented two distinct prediction methodologies namely, Random Forest Regression (RFR) and Long Short-Term Memory (LSTM). Our findings unequivocally favored RFR as the superior predictor for PM2.5 values, achieving accuracy levels of up to 90%.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning approaches; Machine learning algorithms; • Applied computing** → **Health care information systems; Environmental sciences;**

KEYWORDS

AQI, Machine Learning, Data Analysis, Random Forest Regression (RFR), Long Short-Term Memory (LSTM)

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1 INTRODUCTION

Air pollution in Bishkek has become a critical issue, endangering public health. Particularly severe during winter, the city grapples with a thick smog layer, exacerbating respiratory diseases, cancer, strokes, heart attacks, and diabetes [3]. Between 2015 and 2019, respiratory diseases surged by 30.6%, with bronchial asthma cases nearly doubling and lung cancer incidences increasing significantly [3].

Particulate matter (PM), especially PM2.5, plays a pivotal role in air quality degradation, often forming smog. PM2.5, particles smaller than 2.5 micrometers, pose grave health risks. Accurate

prediction of Air Quality Index (AQI) values for PM2.5 is crucial for effective pollution management [11].

Precise AQI prediction models are pivotal for urban development and public health. They empower decision-makers with insights to mitigate pollution's adverse effects and promote sustainable development [9]. This study aims to contribute to this endeavor by enhancing AQI prediction accuracy in Bishkek.

Three prediction methods will be explored: Random Forest Regression (RFR) without lagging variables, RFR with lagging variables, and Long Short-Term Memory (LSTM) models. These models will utilize real-time data to capture air quality fluctuations.

RFR models leverage ensemble learning, while LSTM models specialize in sequential data analysis. By evaluating their performance, this study aims to identify the most effective approach for predicting AQI values in Bishkek. However, limitations such as localized model applicability, limited meteorological data availability, and the need for additional features must be addressed to enhance prediction accuracy and applicability.

2 LITERATURE REVIEW

2.1 Understanding the Air Quality Index (AQI)

The Air Quality Index (AQI) serves as a standardized metric for assessing and reporting air quality conditions. It quantifies the level of pollutants such as PM2.5, PM10, ozone (O3), carbon monoxide (CO), sulfur dioxide (SO2), and nitrogen dioxide (NO2), all of which have significant health impacts. The EPA's AQI calculation method uses a scale ranging from 0 to 500, divided into six categories: Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and Hazardous. This scale provides a clear and easily interpretable depiction of air quality conditions and associated health risks [1].

For PM2.5, the AQI is determined based on either the 24-hour average concentration or the current hourly concentration, whichever is higher. The calculation involves converting pollutant concentrations into a standardized index using predefined breakpoints and a specific formula. This formula considers factors such as the truncated concentration of PM2.5 (Cp), concentration breakpoints (BPHi and BPLo), and the corresponding AQI values (IH_i and IL_o) to produce a reliable measure of air quality. The AQI provides a critical tool for understanding and communicating air quality status, aiding in public health protection and policy-making [1].

2.2 Related Works

Extensive research has been conducted on factors influencing AQI levels and methods for predicting air quality. For example, researchers Isaev, Erkin, et al. conducted a study on AQI levels in Bishkek, developing a model to forecast AQI for the next 24 hours [6]. They collected data on pollutants (NO, CO, NO2, SO2, PM2.5,

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PM10) and meteorological parameters (wind speed and direction, pressure, humidity, temperatures, UV radiation, net radiation, precipitation, wind gust direction) from 2015 to 2020. Testing various prediction methods, including ANN, HPT, XgbR, RFR, KNN, DTR, LaR, and LR, they found that Random Forest Regression (RFR) performed best with a Root Mean Square Error (RMSE) of 28. Parameshachari et al. also evaluated machine learning algorithms for predicting AQI, finding that RFR outperformed Decision Tree Regression (DTR), Linear Regression (LR), and Support Vector Regression (SVR) in predictive accuracy [7]. Building on these studies, our research aims to validate RFR's effectiveness in predicting AQI in different Bishkek locations using comprehensive datasets and rigorous evaluation techniques, contributing to environmental monitoring and decision-making processes.

In addition to RFR, Chang, Yue-Shan, et al. proposed an Aggregated LSTM (ALSTM) model for predicting air pollution [2]. They utilized data from both local and industrial air quality monitoring stations as well as external pollution sources, conducting a comparative analysis with SVR, GBTR, and LSTM models. Their findings demonstrated that ALSTM notably enhanced PM2.5 prediction accuracy across 1–8 hour periods, showcasing its effectiveness in air pollution forecasting. Notably, their study reported an overall RMSE of 3.94 for all regional stations, indicating the ALSTM model's robust performance in capturing air quality variations.

A comprehensive study by Sadriddin et al. analyzed AQI values in Kyrgyzstan compared to 31 other countries, using data from AirNow DOS and the US EPA [8]. They examined factors like population density, tree coverage, and CO2 levels, finding population density had a medium effect on air quality (correlation coefficient of 0.38) and tree coverage had a negative correlation with AQI (-0.43). Seasonal analysis showed significant AQI variation, with worse air quality in Eurasian winters and American summers. Bishkek's AQI varied seasonally, with unhealthy winter levels and good summer levels.

3 RESEARCH METHODOLOGY

3.1 Data collection for training ML model

To develop a robust predictive model for the Air Quality Index (AQI), comprehensive historical data on AQI, humidity, and temperature were collected. IQAir was chosen as the primary source for AQI data due to its reliability, and historical data from five stations in Bishkek was obtained [5]. Meteorological data, including air temperature and relative humidity, was also collected from the same stations. Both Random Forest Regression (RFR) and Long Short-Term Memory (LSTM) models were trained using the same dataset to ensure consistency in model comparison and evaluation.

Real-time data for AQI predictions was retrieved using API requests from open-source providers like The World Air Quality Project [4]. Meteorological forecast data for Bishkek was obtained from Meteoblue via API requests [10].

3.2 Implementation of forecasting methodologies

Various Python libraries, including scikit-learn for Random Forest Regression and TensorFlow for LSTM, were utilized to implement forecasting methods. Scikit-learn offers a comprehensive collection

of machine learning algorithms, while TensorFlow provides flexibility in expressing and executing machine learning models across diverse computing systems.

Data preparation involved gathering AQI and meteorological data, concatenating and cleansing the dataset, and extracting relevant features. Outliers and NaN values were handled to ensure data integrity and consistency, preparing the dataset for further analysis and model training.

Two distinct RFR models were trained: one without lagging variables and another with lagging variables. Each model's implementation included data splitting, model fitting, and feature importance analysis to evaluate the significance of input features in predicting AQI values. Feature importance analysis was conducted using the feature importances attribute of the Random Forest Regression function in Python.

Similar to RFR, the LSTM model underwent data preparation, outlier handling, and data splitting before model fitting. Scaling and reshaping of data enabled effective training of the LSTM model, with emphasis on capturing temporal dependencies and patterns within the data.

A Flask web application was developed to display predicted AQI values for multiple stations in Bishkek. Real-time data retrieval, model loading, and prediction processes were implemented, providing users with an intuitive interface to access and interact with predictive models for informed decision-making.

4 RESULTS

4.0.1 Variable importance. The analysis of variable importance for the four types of prediction models using Random Forest Regression (RFR) provides valuable insights into the factors influencing air quality predictions. The importance of each feature varies depending on the model configuration, as shown in Figure 1.

For hourly predictions without a lagging variable, temperature emerges as the most important feature, indicating its significant impact on predicting AQI values for short-term forecasts. However, when incorporating a lagging variable for hourly predictions, the previous AQI value becomes the most important feature, suggesting the importance of temporal dependencies in predicting current air quality levels.

In daily predictions without a lagging variable, temperature remains crucial, with even higher importance compared to hourly predictions. This indicates the stronger influence of temperature on shaping daily variations in air quality. Conversely, daily predictions with a lagging variable emphasize the importance of the previous day's AQI value, underscoring the persistence of air quality conditions over time.

Overall, the analysis highlights the dynamic nature of variable importance in air quality prediction models and underscores the significance of incorporating both meteorological factors and historical AQI data for accurate forecasts.

4.0.2 Evaluation of the prediction methods. The evaluation of prediction methods reveals varying levels of accuracy and performance across different models. For hourly predictions, the RFR model with a lagging variable outperforms the model without a lagging variable, achieving higher accuracy (89% vs. 82%) and lower RMSE (24.8 vs. 31.5). The inclusion of lagging variables enables the model

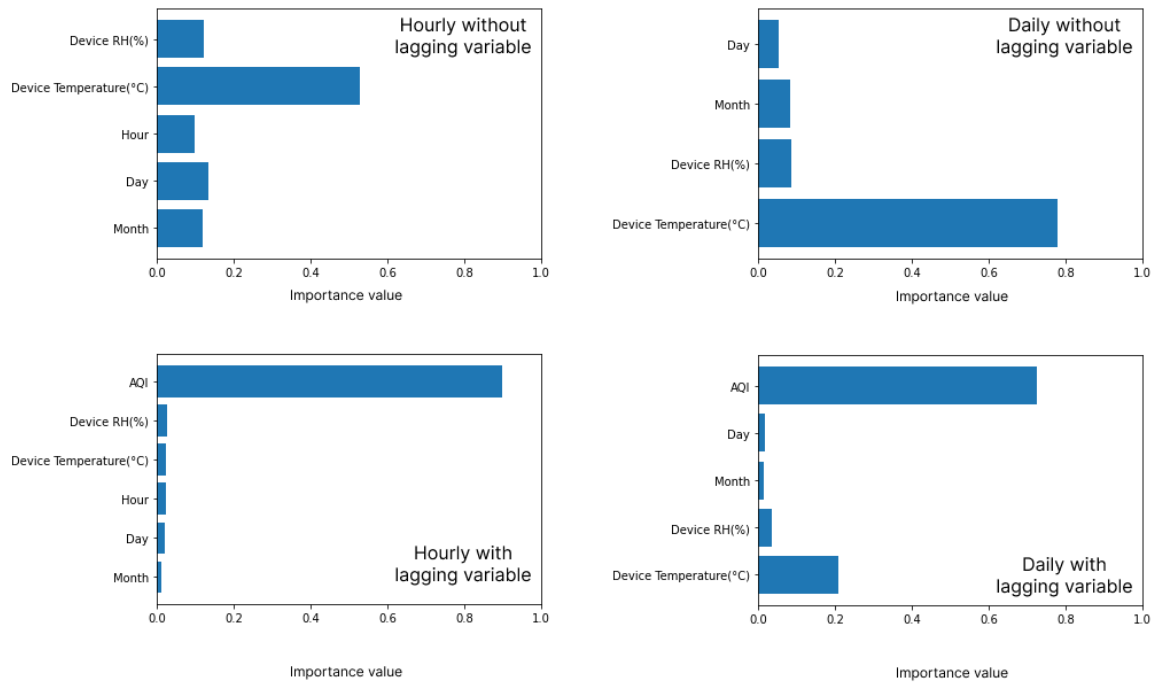


Figure 1: Importance of variables in four different prediction models.

to capture temporal dependencies more effectively, resulting in improved predictions, as illustrated in Figures 2a and 2b.

Similarly, for daily predictions, incorporating lagging variables leads to enhanced performance, with the RFR model achieving higher accuracy (86% vs. 78%) and lower RMSE (27.3 vs. 33.1) compared to the model without lagging variables. Figures 3a and 3b depict the improved alignment between predicted and real values when lagging variables are included.

In contrast, the LSTM model demonstrates inferior performance, with higher RMSE values for both hourly (44) and daily (40) predictions. Figures 4a and 4b illustrate the discrepancies between predicted and real values, highlighting the challenges of accurately capturing air quality fluctuations using LSTM.

These results emphasize the effectiveness of Random Forest Regression, particularly when incorporating lagging variables, for air quality prediction tasks. The RFR models demonstrate superior performance in capturing both short-term and long-term variations in AQI compared to LSTM.

4.1 Integration of Prediction Models

To facilitate user access to air quality predictions, we developed a web application that offers both hourly and daily forecasts using Random Forest Regression models. The web application provides users with intuitive interfaces to explore predicted AQI values for different stations in Bishkek, as depicted in Figure 5. By presenting

predictions from both models side by side, users can compare the impact of incorporating lagging variables on forecast accuracy.

Additionally, we implemented a geographical heatmap to visualize the spatial distribution of AQI values across Bishkek, as shown in Figure 6. This heatmap provides users with a comprehensive overview of air quality conditions across different regions, facilitating informed decision-making and risk assessment.

As part of future development plans, we aim to integrate personalized notifications and historical data analysis features into the web application to enhance user engagement and provide additional insights into air quality trends over time.

Furthermore, we are actively working on the development of a mobile application dedicated to predicting AQI values. The mobile app will offer real-time updates, personalized predictions based on user location, and historical data analysis capabilities to empower users in managing their exposure to air pollution and making informed decisions to protect their health and well-being.

By integrating prediction models into user-friendly applications, we aim to democratize access to air quality forecasts and empower individuals to take proactive measures to mitigate the adverse effects of poor air quality on their health and environment.

5 CONCLUSION

In conclusion, this article aimed to investigate and develop models for predicting air quality using various techniques, including Random Forest Regression (RFR) and Long Short-Term Memory

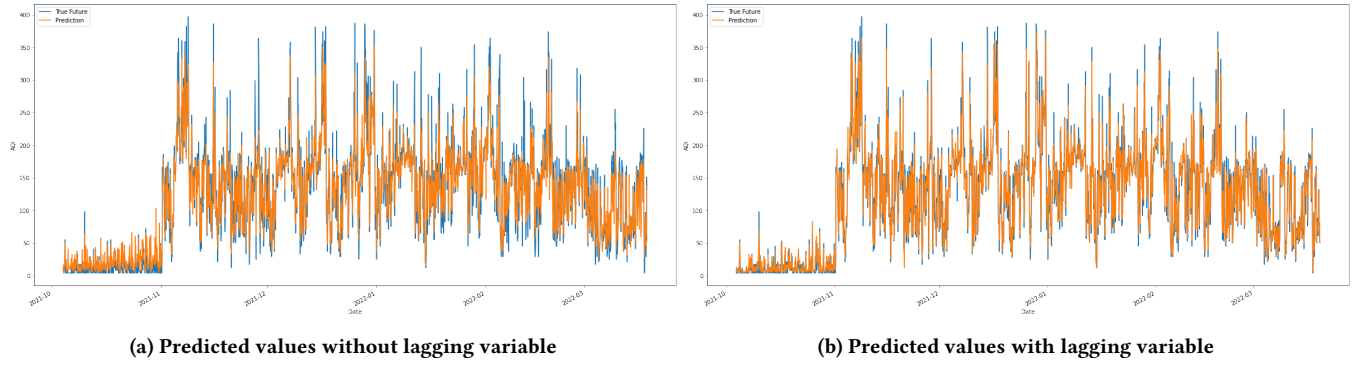


Figure 2: Graphs of predicted hourly AQI values on test data using RFR

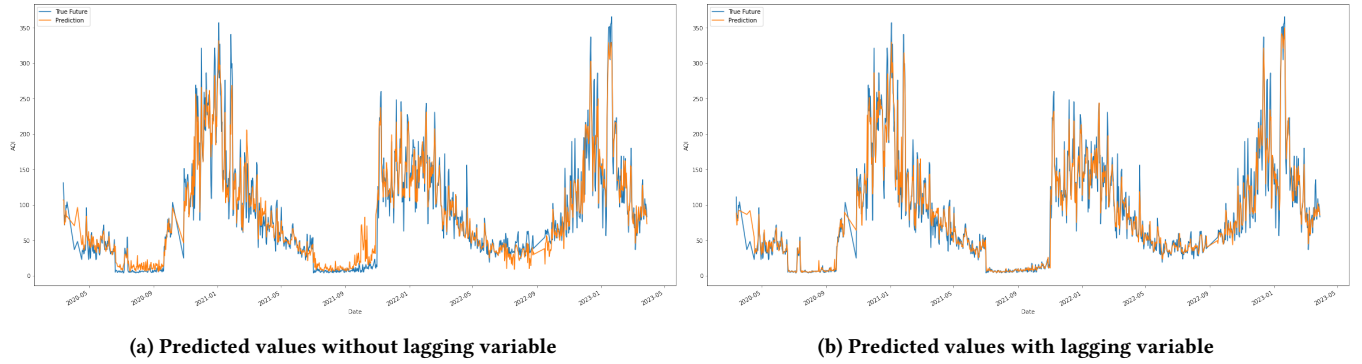


Figure 3: Graphs of predicted daily AQI values on test data using RFR

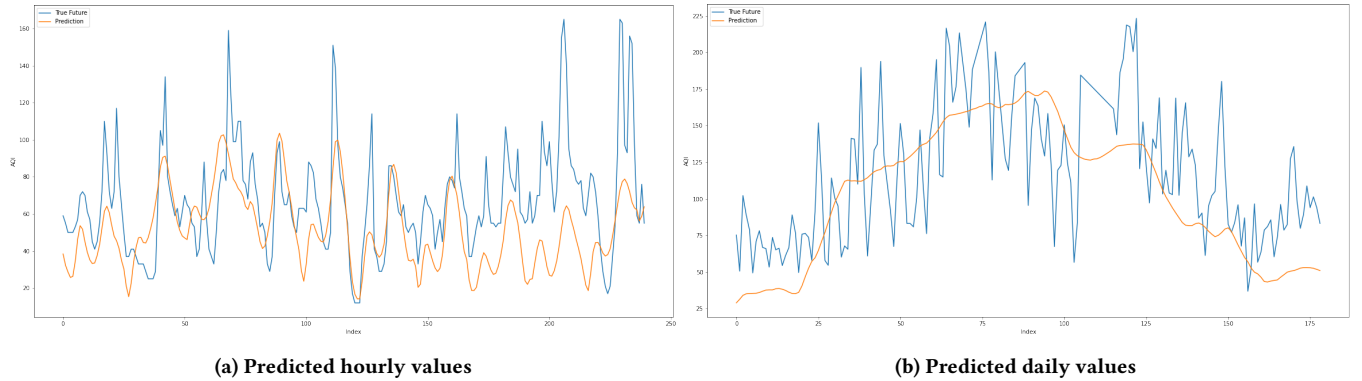


Figure 4: Graphs of predicted AQI values on test data using LSTM

(LSTM). Through extensive data analysis and experimentation, we have made significant progress in understanding the factors that influence air quality and predicting Air Quality Index (AQI) values.

Our results demonstrate that RFR models, both with and without lagging variables, offer valuable insights into the significant features impacting air quality predictions. The variable importance analysis revealed that temperature played a crucial role in shaping air quality variations, with its impact differing between hourly and daily predictions.

Furthermore, our evaluation of prediction accuracy showed that RFR models achieved moderate levels of accuracy, with hourly predictions achieving 82% accuracy and daily predictions achieving 78% accuracy. The inclusion of a lagging variable in the models improved their performance, with an accuracy of 89% for hourly predictions and 86% for daily predictions. When considering the root mean square error (RMSE) as a measure of prediction error, the RFR model without a lagging variable achieved an RMSE of 31.5 for hourly predictions and 33.1 for daily predictions. The addition of a

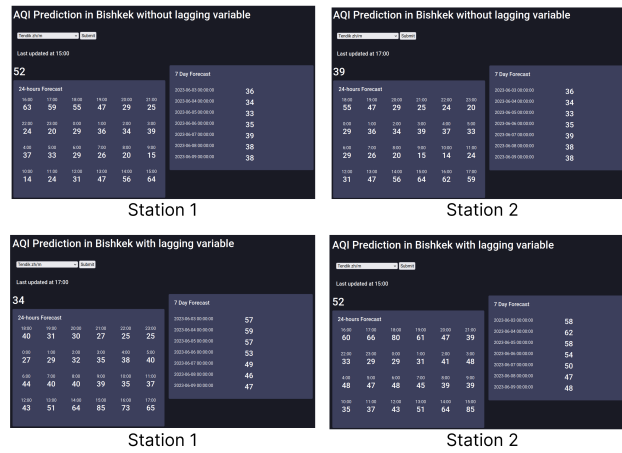


Figure 5: The web application. Station1 - Tendik zh/m, Station 2 - Bishkek International School

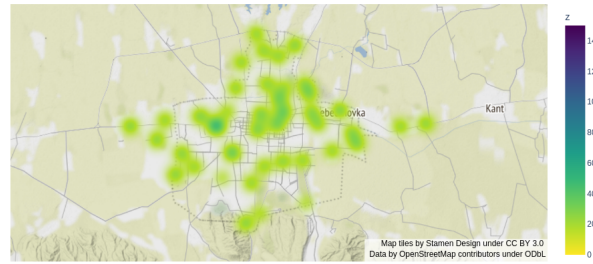


Figure 6: Geographical heatmap of AQI values in Bishkek

lagging variable led to a reduction in RMSE, resulting in values of 24.8 for hourly predictions and 27.3 for daily predictions, surpassing the RMSE reported by Isaev, Erkin et al., which stood at 28. These RMSE values indicate the average magnitude of the prediction errors, with lower values indicating better alignment between the predicted and actual AQI values.

On the contrary, the LSTM model exhibited inferior accuracy and higher RMSE values when compared to the RFR models. Specifically, for hourly predictions, the LSTM model yielded an RMSE of 44, indicative of significant prediction errors and noticeable deviations from the actual values. This suggests considerable challenges in capturing the true patterns of air quality fluctuations. Similarly, for daily predictions, the LSTM model attained an RMSE of 40, reflecting a notable prediction error and an inability to accurately capture the actual pattern of air quality variations. These outcomes starkly contrast with those achieved by Chang, Yue-Shan et al., who reported an RMSE of 3.94, indicating the substantially poorer performance of our LSTM model in comparison.

These findings suggest that the RFR models outperformed the LSTM model in accurately predicting both hourly and daily AQI

values. The RFR models exhibited lower RMSE values and demonstrated better alignment with the real AQI values, particularly during peak points.

In addition to model development, we have also developed web applications that provide users with access to the predicted AQI values. The web apps incorporated both RFR and LSTM models, allowing users to compare predictions from different techniques for different stations. The geographical heatmap visualization showcased the distribution of stations throughout Bishkek, providing a comprehensive view of air quality across the city.

Looking ahead, we have outlined future directions for this research. These include the development of a mobile application that offers a more convenient and user-friendly platform for AQI prediction. The mobile app will leverage advanced machine learning algorithms and real-time data to provide personalized predictions and empower users to make informed decisions based on air quality conditions.

Overall, this research work contributes to the field of air quality prediction by offering insights into significant factors influencing air quality and developing models that aid in predicting AQI values. By understanding and forecasting air quality, we can take proactive measures to protect public health, make informed decisions, and

implement effective strategies to mitigate the impact of air pollution on the environment and human well-being.

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